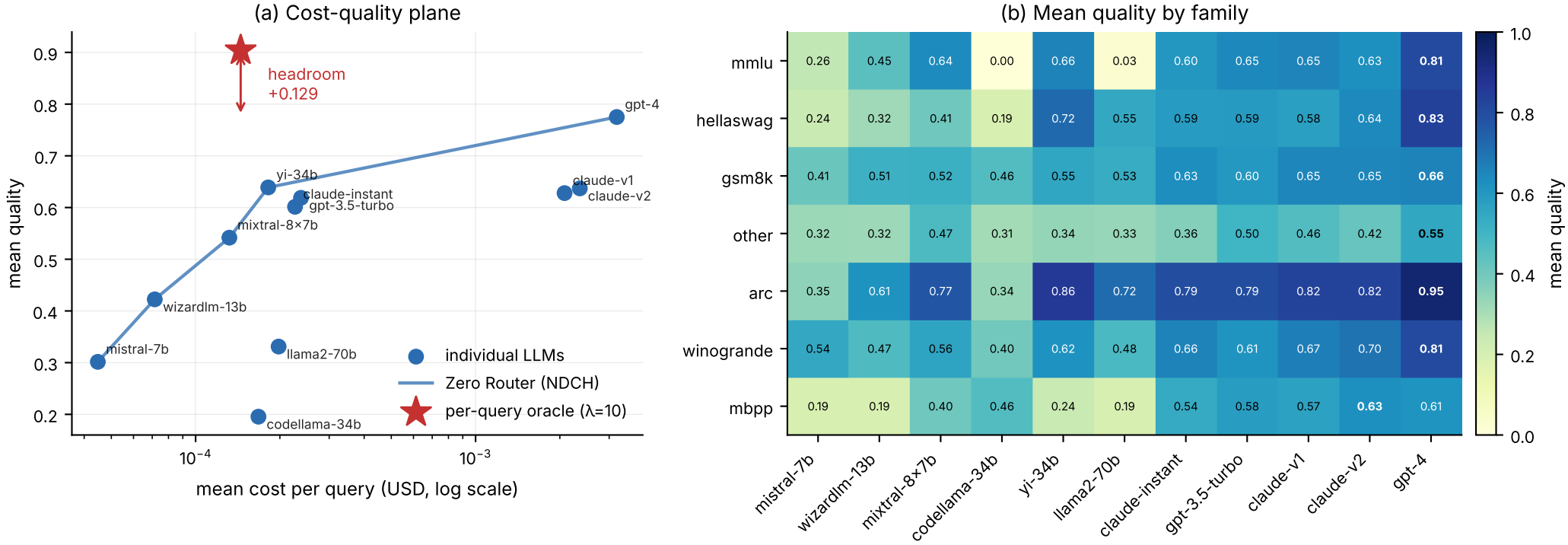
(a) Cost-quality plane
mean quality
mean cost per query (USD, log scale)
headroom
+0.129
gpt-4
yi-34b
claude-v1
claude-instant
claude-v2
gpt-3.5-turbo
mixtral-8×7b
wizardlm-13b
mistral-7b
llama2-70b
codellama-34b
individual LLMs
Zero Router (NDCH)
per-query oracle (λ=10)
(b) Mean quality by family
mmlu 0.26 0.45 0.64 0.00 0.66 0.03 0.60 0.65 0.65 0.63 0.81
hellaswag 0.24 0.32 0.41 0.19 0.72 0.55 0.59 0.59 0.58 0.64 0.83
gsm8k 0.41 0.51 0.52 0.46 0.55 0.53 0.63 0.60 0.65 0.65 0.66
other 0.32 0.32 0.47 0.31 0.34 0.33 0.36 0.50 0.46 0.42 0.55
arc 0.35 0.61 0.77 0.34 0.86 0.72 0.79 0.79 0.82 0.82 0.95
winogrande 0.54 0.47 0.56 0.40 0.62 0.48 0.66 0.61 0.67 0.70 0.81
mbpp 0.19 0.19 0.40 0.46 0.24 0.19 0.54 0.58 0.57 0.63 0.61
mistral-7b
wizardlm-13b
mixtral-8×7b
codellama-34b
yi-34b
llama2-70b
claude-instant
gpt-3.5-turbo
claude-v1
claude-v2
gpt-4
mean quality